

Kyle Simonetti

CS-499

March 29, 2026

Milestone 3

The artifact I have chosen is the inventory app I made using android studio for CS-360: mobile architecture and programming. This android application allows users to log in to the application using a username and password, which brings them to a screen to track inventory items. Users can add, delete, and change items as they see fit, with notifications sent to their phone when inventory is low. This application was finalized in October of 2025.

I have chosen this artifact to enhance as, firstly, it is the project I made that I was most proud of because of its demonstration of a variety of skills. I also picked this artifact because it showcases my skills in making android applications, which aligns with the career path I am moving towards, that being mobile application development, and it aligns with my current job, this application being something that could enhance our current inventory management. The specific components that showcase my skills in algorithms and data structure is the addition of a HashMap. This data structure makes looking up inventory from $O(n)$ to $O(1)$, which will be incredibly useful when inventory gets large. I have met course outcome three, as I was able to use algorithmic principles to solve the problem of efficiency in looking up inventory. Not only is looking up an item more efficient, but adding, updating, and deleting items also become more efficient. The application can also quickly discern if an item exists in the database, and add or update the item accordingly without having to iterate through the inventory.

As I was creating this enhancement, I had issues changing a lot of my code to work with the new HashMap, as there was a lot to change and it was easy to forget what still needed to be reworked. I also had to change my repository to a singleton, as after changing to use a HashMap, I found out on creating the repository each time, it created two separate HashMaps in my InventoryActivity and my ItemDetailsActivity since each activity created its own repository.